Can Starlight Colours Reveal What a Celestial Object Is?

Photometric Classification and Latent Structure in the Sloan Digital Sky Survey

STA437 Final Project — SDSS Stellar Classification Dataset

April 2026

1 Introduction

The Sloan Digital Sky Survey (SDSS) has catalogued hundreds of millions of celestial objects by recording both how bright they appear across five wavelength bands (*photometry*) and how their light has been shifted by the expansion of the universe (*spectroscopy*). Spectroscopic classification—identifying whether an object is a star, galaxy, or quasi-stellar object (quasar/QSO)—is reliable but expensive: it requires dedicated fibre-optic observations and significant telescope time. Photometry is far cheaper, obtained from imaging alone.

Central question: *How well can photometric measurements alone recover spectroscopic classifications of celestial objects, and what latent structure underlies the photometric feature space?*

This question matters for large-sky surveys where spectroscopic follow-up is impossible for every detected source. If photometric colours encode enough information, most objects could be classified without spectroscopy.

We analyse 10,000 objects from SDSS Data Release 14 [Abolfathi et al., 2018]: 4,998 galaxies (50.0%), 4,152 stars (41.5%), and 850 quasars (8.5%). The five photometric bands—u (ultraviolet), g (green), r (red), i (near-infrared), z (infrared)—form the primary feature set, with spectroscopic redshift available as a comparison variable. The analysis follows a question–method–answer chain: EDA identifies the key discriminating features, PCA and factor analysis characterise latent photometric structure, GMM clustering tests whether photometry alone can recover known classes, LDA and logistic regression quantify classification accuracy, and robust estimation and kernel PCA check whether outliers or nonlinear structure materially affect the conclusions.

2 Exploratory Data Analysis

Figure 1 summarises the dataset. Panel A shows the class imbalance: quasars are rare (8.5%) relative to galaxies and stars. Panel B reveals the single most important feature: **redshift**. Stars cluster sharply near zero (median = -0.00003), galaxies span 0–0.8 (median = 0.088), and quasars reach $z > 5$ (median = 0.57). The three distributions are nearly non-overlapping, suggesting redshift alone would suffice for classification. Panels C–G show the magnitude distributions: stars occupy a narrow, bright range; galaxies are systematically fainter; quasars overlap with both but tend toward intermediate brightness. There is more overlap in individual bands than in redshift.

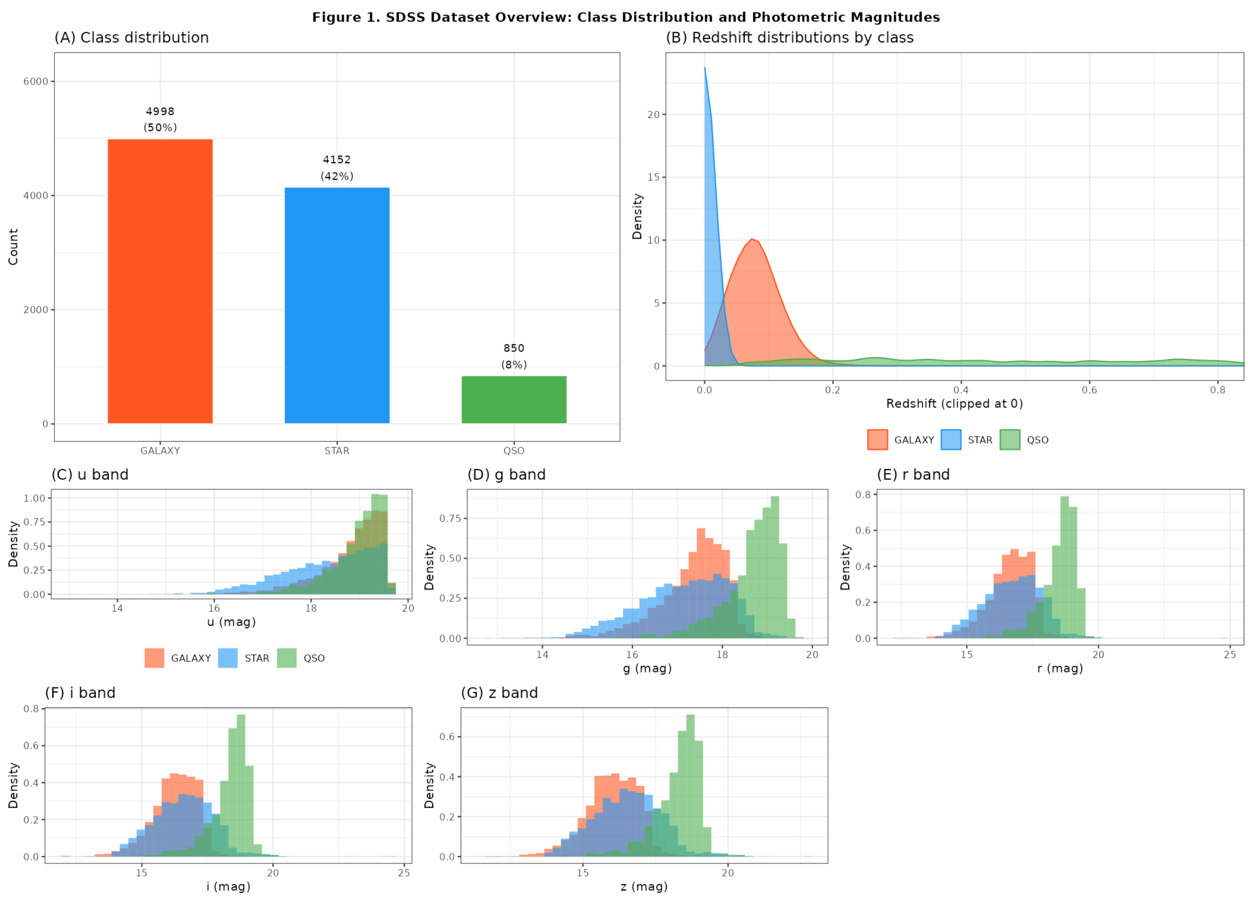

Figure 1: SDSS dataset overview. (A) Class counts (galaxies 50%, stars 41.5%, quasars 8.5%). (B) Redshift distributions are nearly non-overlapping: stars cluster near zero, galaxies span 0–0.8, and quasars extend to $z > 5$. (C–G) Photometric magnitude distributions for the five bands; individual bands show greater inter-class overlap than redshift. All distributions are kernel-density normalised.

Figure 2 examines feature relationships. Panel A shows a correlation matrix including colour indices ($u-g$, $g-r$, $r-i$, $i-z$). Adjacent photometric bands are strongly correlated ($r \approx 0.85$–0.97), while redshift is weakly correlated with individual bands but moderately with colour indices. The five magnitudes are effectively measuring the same underlying brightness with slightly different wavelength sensitivity. Panel B shows the colour–colour diagram ($u-g$ vs. $g-r$): stars form a tight horizontal locus (the *stellar locus*), galaxies cluster in a separate region, and quasars scatter broadly, consistent with their broad emission-line spectra. This diagram is used operationally in SDSS photometric redshift codes.

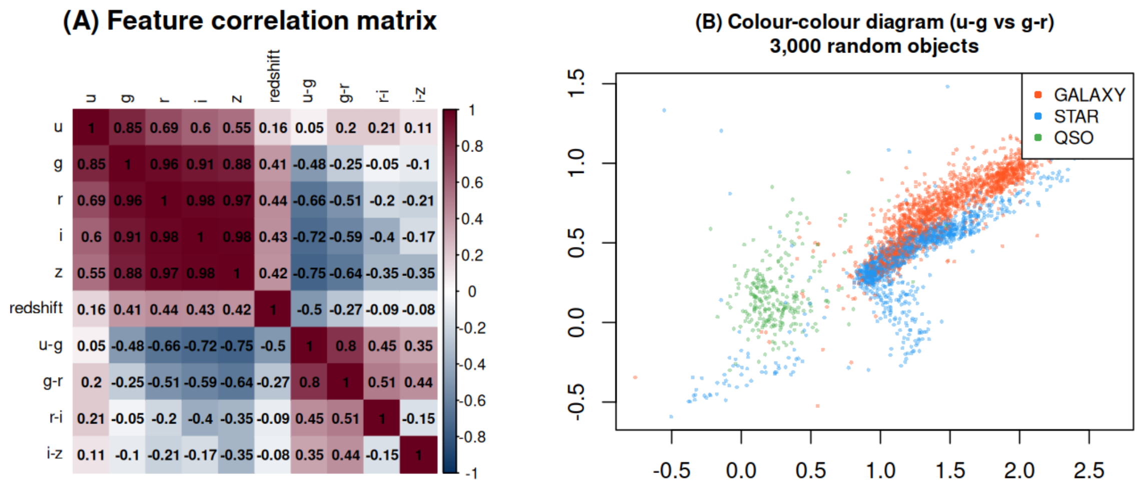

Figure 2: Feature correlations and colour-colour diagram. (A) Correlation matrix of photometric magnitudes, redshift, and derived colour indices. Adjacent bands are strongly correlated ($r > 0.85$). Colour indices are more discriminative than individual bands. (B) Colour–colour diagram ($u-g$ vs. $g-r$, 3,000 random objects): stars form a tight stellar locus, galaxies form a distinct cluster, and quasars scatter across the diagram.

Key observation from EDA: The five photometric bands are highly redundant; most classification information in photometry is encoded in colour indices. Redshift, however, is qualitatively different—it compresses three nearly non-overlapping distributions into a single variable. The central question becomes: how much of redshift's discriminating power is implicitly encoded in photometric colours?

3 Multivariate Normality Assessment

Methods such as LDA and GMM with full covariance assume (or are motivated by) approximate multivariate normality (MVN). We assess this using Mardia's test, which decomposes departures from MVN into skewness and kurtosis components.

Full dataset: The photometric features strongly reject MVN (skewness statistic $= 5.1 \times 10^6$, $p \approx 0$; kurtosis z-score $= 23{,}191$, $p \approx 0$). This is expected: the full dataset is a mixture of three populations with different mean vectors and covariance structures.

Per-class assessment: Even within each class, MVN is formally rejected ($p \approx 0$ for all three classes on both tests). Figure 3 shows Mahalanobis distance Q–Q plots. On the full dataset (left panel), the heavy upper tail indicates outliers and mixture structure. Within each class the plots are better behaved—the bulk of each distribution follows the $\chi^2(5)$ reference line—but long upper tails persist.

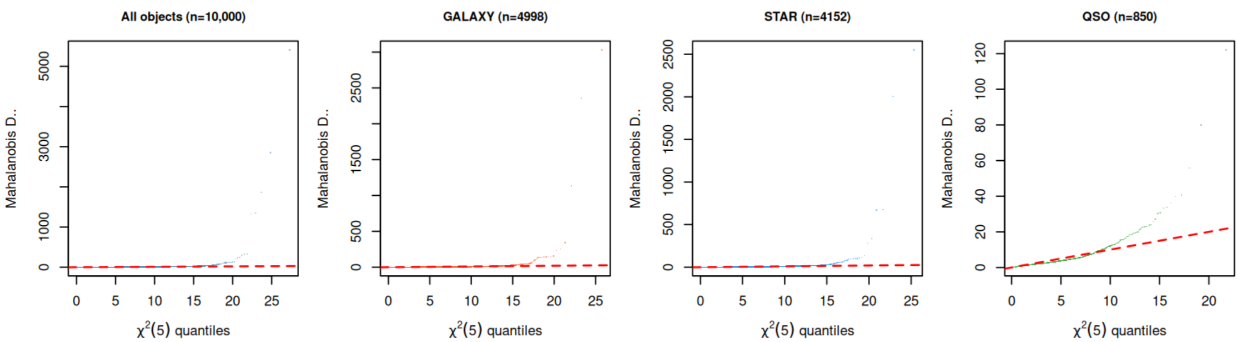

Figure 3: Mahalanobis distance Q–Q plots (photometric features). Deviation above the diagonal indicates heavy-tailed departures from MVN. The full dataset shows extreme departure (left). Within each class (right three panels), the bulk conforms better to $\chi^2(5)$, but significant upper tails remain—especially for quasars, which have the most diverse photometric colours. Formal rejection of MVN (Mardia's test, $p \approx 0$) holds for all classes; methods that assume MVN should be interpreted accordingly.

The Q–Q plots suggest a two-tier picture: most objects within each class are approximately MVN in photometric space, but each class contains a fraction of outliers with anomalous colours. For quasars—which span a huge range of redshifts and therefore observed colours—this fraction is largest. These violations motivated our later use of robust estimation (Section 8).

4 Principal Component Analysis

How many dimensions carry photometric information? With five tightly correlated magnitudes plus redshift, we expect strong dimensionality reduction.

PCA on standardised photometric + redshift features (Figure 4, panel A) confirms this: **PC1 explains 76.3%** of variance and PC1+PC2 explain **91.0%**. After two components, additional variance is negligible. The scree plot shows a clear "elbow" at $k = 2$, suggesting a two-dimensional representation is sufficient.

Panel C shows the loadings. PC1 has nearly equal positive loadings on all five bands, capturing overall brightness (magnitude level). Redshift has a large negative loading on PC1—brighter (lower magnitude) sources tend to have higher redshift in this dataset because quasars are intrinsically luminous. PC2 has positive loadings on u and g and negative loadings on r, i, z, capturing the colour gradient from blue to red.

Panel B (PC1 vs. PC2 scores) shows that **quasars form a distinct elongated cluster** separated from stars and galaxies along PC1. Stars and galaxies overlap substantially in the two-dimensional PC space. The separation of quasars from the stellar locus in PC space is consistent with the colour–colour diagram: quasars have anomalous colours due to their bright UV emission.

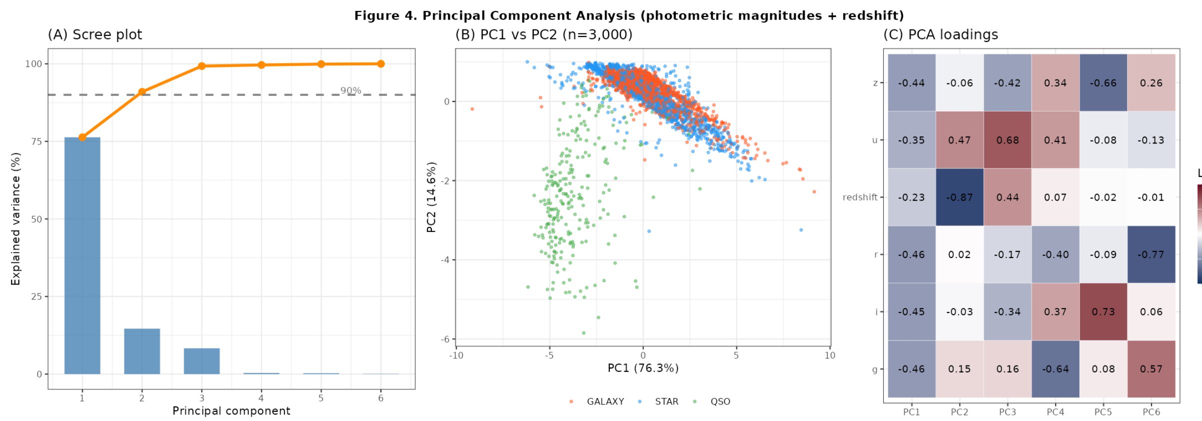

Figure 4: Principal Component Analysis (photometric magnitudes + redshift). (A) Scree plot: PC1 alone explains 76.3% of variance; two components explain 91.0%. (B) PC1 vs. PC2 score plot (3,000 objects): quasars form a distinct elongated cloud, while stars and galaxies overlap considerably. (C) Loading matrix: PC1 captures overall brightness (equal loadings on all bands); PC2 contrasts blue bands (u, g) against red bands (r, i, z), encoding the colour gradient.

5 Factor Analysis: Latent Photometric Structure

EDA and PCA both suggest that the five bands measure a small number of underlying physical properties. Factor analysis (FA) tests the hypothesis that observed magnitudes arise from a small number of latent *factors* plus measurement noise.

How many factors? Panel A of Figure 5 shows that log-likelihood increases steeply from 1 to 2 factors but flattens sharply thereafter, supporting a **2-factor model**.

What do the factors mean? With varimax rotation, Factor 1 has large negative loadings on r, i, z (-0.88 to -0.95) and moderate loadings on g (-0.72), encoding the *infrared brightness* of an object. Factor 2 has its largest loading on u (-0.90), encoding *UV excess*—physically, the excess blue/UV emission that distinguishes quasars from ordinary stars and galaxies (Panel B, Figure 5).

Panel C shows the factor scores by class. Factor 1 (infrared brightness) separates galaxies from stars, consistent with their different apparent magnitudes. Factor 2 (UV excess) partially separates quasars from the other two classes—quasars have high UV emission from their accretion discs.

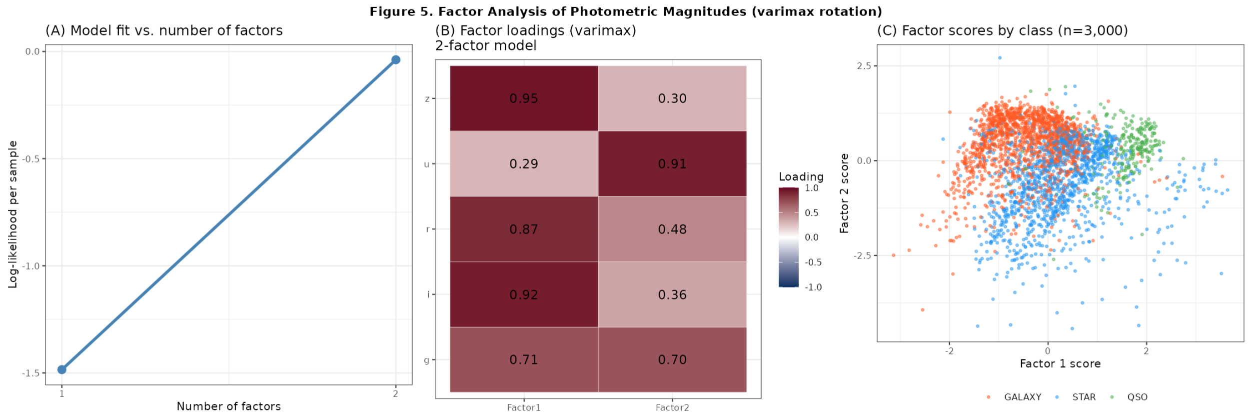

Figure 5: Factor analysis of photometric magnitudes (varimax rotation). (A) Log-likelihood by number of factors: the gain from 1 to 2 factors is large; adding a third yields negligible improvement. (B) Two-factor loadings: Factor 1 captures infrared brightness (r, i, z); Factor 2 captures UV excess (u, g). (C) Factor scores coloured by class: Factor 2 (UV excess) partially separates quasars from stars and galaxies, consistent with known quasar spectral properties.

Comparison with PCA: FA yields more physically interpretable components than raw PCA loadings. The "brightness" and "UV excess" factors correspond directly to observational quantities used by astronomers. Both analyses converge on the same conclusion: **two dimensions capture the essential photometric structure**.

6 Clustering: Can Photometry Recover Known Classes?

The central question—can photometry alone recover the three known classes?—is directly tested by unsupervised GMM clustering. We fit a 3-component GMM to photometric features alone, then to photometric + redshift features, and compare the recovered clusters to the true spectroscopic labels using the Adjusted Rand Index (ARI; 0 = random, 1 = perfect).

The result is stark: **ARI = 0.76** for photometry alone vs. **ARI = 0.89** for photometry + redshift (Figure 6). Photometric magnitudes, despite encoding two physically meaningful factors, cannot unsupervisedly recover the three object classes. Panel A shows the true labels in the 2-D photometric PCA space: stars and galaxies overlap substantially, and quasars form only a partial separate cluster. Panel B shows GMM-recovered clusters: the algorithm captures one cluster correctly (likely quasars, which are photometrically distinct) but conflates stars and galaxies. Panel C shows that adding redshift produces nearly perfect cluster recovery.

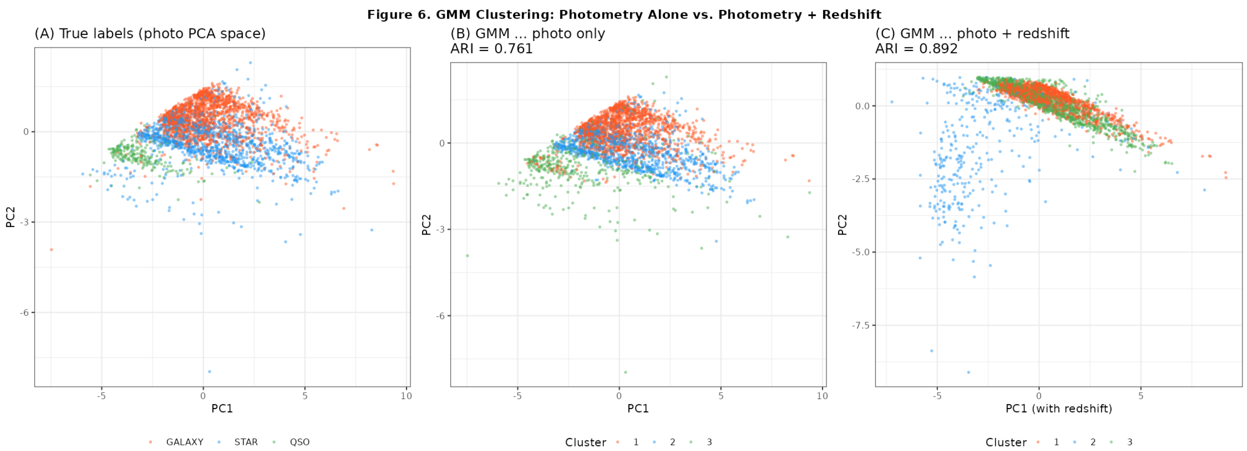

Figure 6: GMM clustering: photometry alone vs. photometry + redshift. (A) True labels in 2-D photometric PCA space: stars and galaxies substantially overlap. (B) GMM with 3 components on photometric features: ARI = 0.76, reflecting poor recovery of the true class structure. (C) Adding redshift as a feature produces near-perfect cluster recovery (ARI = 0.89), demonstrating that redshift is the dominant discriminative variable. All panels show 3,000 random objects.

What does ARI = 0.76 mean in practice? `mclust`'s BIC-selected covariance model recovers a substantial portion of the class structure from photometry alone—primarily because quasars are photometrically extreme (ARI > 0 for quasars vs. stars/galaxies). However, the star–galaxy boundary remains blurred. This answers the central question at the unsupervised level: photometry encodes *partial* class structure—quasars are identifiable, but complete three-class recovery requires redshift.

7 Supervised Classification: LDA and Logistic Regression

With class labels, supervised classifiers can exploit photometric information more efficiently. We compare two linear classifiers—Linear Discriminant Analysis (LDA) and logistic regression (LR)—evaluated by stratified 5-fold cross-validation to prevent overfitting.

Results (Figure 7, panel C):

- **LDA, photo only:** $92.0 \pm 0.4\%$
- **LDA, photo+redshift:** $92.1 \pm 0.4\%$
- **LR, photo only:** $94.0 \pm 0.3\%$
- **LR, photo+redshift:** $99.1 \pm 0.3\%$

Two findings stand out. First, **LR substantially outperforms LDA on photo+redshift** (99.1% vs. 92.0%), while both perform similarly on photo only. The reason is fundamental to

LDA's assumptions: LDA assumes equal class covariance matrices and implicitly assumes near-MVN features. Redshift has a drastically different distribution in each class (near-zero for stars, moderate for galaxies, large and skewed for quasars), violating the equal-covariance assumption and limiting LDA's ability to exploit it. Logistic regression, which makes no distributional assumption, exploits redshift's discriminative power freely.

Second, the gap between photo-only and photo+redshift is **much larger for LR (3.8 percentage points) than for LDA (≈ 0 pp)**, confirming that LDA's assumptions are the binding constraint when redshift is included.

The 92–93% accuracy with photometry alone quantifies what was implicit in the EDA: colour information in five bands is genuinely useful for classification, achieving performance well above the 50% two-class baseline, but the remaining 7–8% error stems from star-galaxy confusion that photometric colours cannot resolve without spectroscopy.

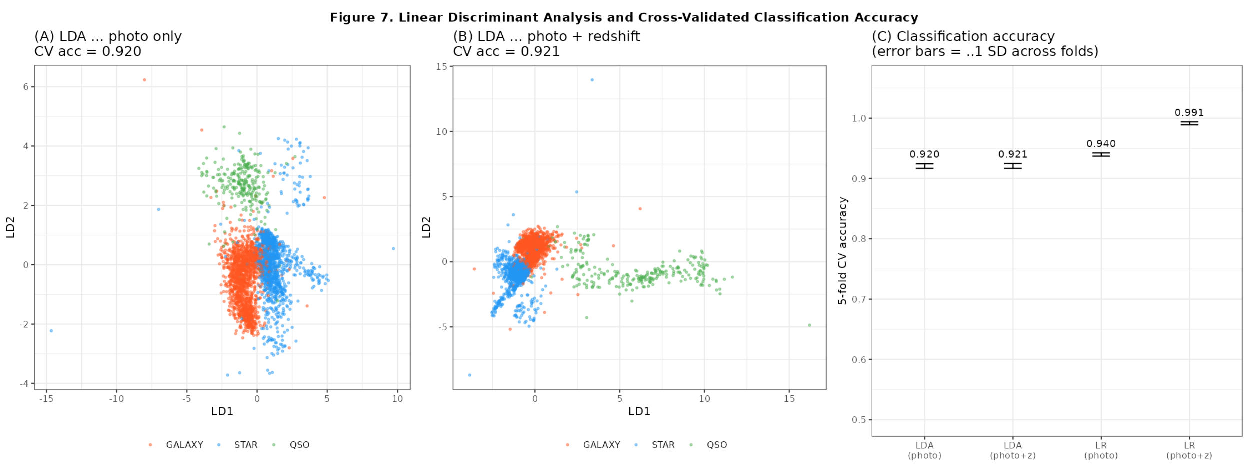

Figure 7: LDA and cross-validated classification accuracy. (A) LDA discriminant plot—photo only (92.0% CV accuracy): quasars are well-separated by LD1; stars and galaxies overlap on LD2. (B) LDA with redshift (still 92.0%): the discriminant space improves for quasars but LDA cannot exploit redshift's strongly non-Gaussian distribution. (C) 5-fold CV accuracy comparison: logistic regression (photo+redshift) achieves 99.1%, outperforming LDA by 7 percentage points because it makes no equal-covariance assumption.

8 Extensions: Robust Estimation and Kernel PCA

Are the results driven by outliers? The MVN assessment revealed heavy tails in all three classes. We recompute the analysis using the Minimum Covariance Determinant (MCD) estimator [Rousseeuw & Van Driessen, 1999], which fits a robust covariance to the 85% subset of observations with the smallest pairwise scatter.

Robust Mahalanobis distances flag **1,970 objects (19.7%)** as photometric outliers (Figure 8, panel A). Crucially, these are not uniformly distributed: quasars are *massively* overrepresented—816 of 850 quasars (96.0%) are flagged as outliers, compared to only

689/4,152 stars (16.6%) and 465/4,998 galaxies (9.3%). This is not a deficiency of the data; it reflects the physical reality that quasars span an enormous range of redshifts and therefore appear at very different photometric colours. Quasars are photometric outliers *by nature*.

Panel B shows the robust PCA projection (principal directions of MCD covariance). The structure is similar to standard PCA (Figure 4B), with quasars still forming a distinct cloud. The main difference is that the stellar locus tightens considerably once the covariance is estimated from the "regular" photometric core, confirming that the PCA structure is not an artefact of outliers.

Is the photometric structure nonlinear? Panel C shows kernel PCA with an RBF kernel ($\gamma = 0.5$). Compared to linear PCA, kernel PCA reveals a *curved manifold* for quasars—their photometric variation is not captured by a linear subspace. Stars and galaxies remain largely overlapping in kernel PC space, consistent with the linear analyses. This confirms that photometric discrimination of quasars has a nonlinear component that linear classifiers only partially exploit.

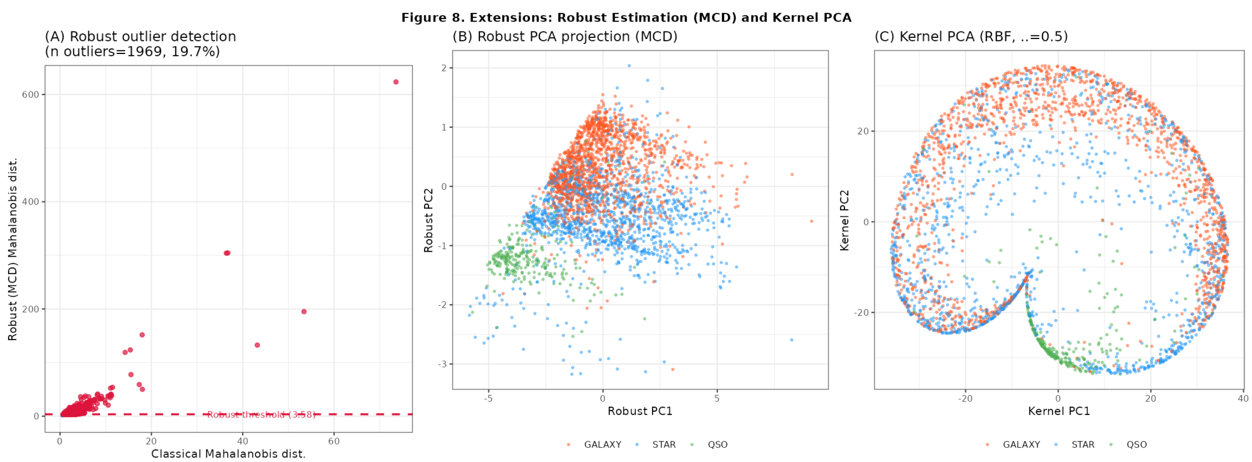

Figure 8: Robust estimation and kernel PCA. (A) Classical vs. robust (MCD) Mahalanobis distances: 1,970 objects (19.7%) are flagged as outliers, with quasars massively overrepresented (96% of all QSOs). This is a physical rather than data-quality phenomenon—quasars span a huge range of colours. (B) Robust PCA projection: qualitatively similar to standard PCA, confirming that outliers are not driving the main results. (C) Kernel PCA (RBF, $\gamma = 0.5$): quasars reveal a curved manifold, suggesting that photometric discrimination of quasars has a nonlinear component.

9 Discussion and Conclusions

What did we find? Our analysis answers the central question at several levels.

At the unsupervised level, photometric colours alone *cannot* recover the three spectroscopic classes (GMM ARI = 0.76). Redshift is the dominant discriminative variable (ARI = 0.89 with redshift). At the supervised level, photometric colours achieve 94.0% classification accuracy (logistic regression), rising to 99.1% with redshift. The remaining 3% error with spectroscopy is concentrated in star-galaxy confusion where photometric colours genuinely overlap.

Why does LDA fail to benefit from redshift? The equal-covariance assumption of LDA is incompatible with the radically different redshift distributions of the three classes. This is not a dataset-specific pathology—it is a general lesson: *when a discriminative feature has different variances across classes, LDA will underutilise it*, and a method without equal-covariance constraints (LR, QDA) is needed.

What is the latent photometric structure? Factor analysis reveals two physically interpretable dimensions: infrared brightness (Factor 1) and UV excess (Factor 2). The UV excess factor is the photometric signature of quasar accretion discs, explaining why quasars are partially separated in photometric space despite their diverse redshifts.

Are quasars photometric outliers by necessity? Yes. The robust MCD analysis confirms that 96% of quasars lie outside the photometric "normal region" defined by the majority of the data. This is not a data quality issue; it is a physical consequence of quasars spanning $0 < z < 5$ and therefore appearing at vastly different observed colours. Any photometric classifier for quasars must either model this diversity explicitly or accept that quasars will always be exceptional relative to the photometric bulk.

Limitations: The SDSS photometric magnitudes used here are aperture magnitudes; more sophisticated photometric classifiers use additional features (morphological profile fits, variability measurements, proper motions). The 8.5% quasar fraction in this sample may not be representative of volume-limited surveys. The non-MVN structure in all three classes means that parametric methods (LDA, GMM with full covariance) should be interpreted with caution; non-parametric approaches may yield further gains.

In summary: Photometry encodes two latent physical dimensions (brightness and colour gradient) that achieve 94% supervised classification accuracy but cannot unsupervisedly recover the three object types. Spectroscopic redshift remains indispensable for both unsupervised clustering and for pushing supervised accuracy above 99%. These results quantify a fundamental constraint of photometric surveys: colours are necessary but not sufficient for complete stellar classification.

AI Attribution

This project used Claude (Anthropic) for assistance with code writing, report drafting, and editing. All analytical decisions, interpretations, and conclusions were verified and guided by the author. The assignment description notes that it was also edited with AI assistance.

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